



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**
WEEKLY PROGRESS REPORT (01/06/2026 – 07/06/2026)

WEEK 3

PROJECT NAME

PRITHVI EO MODELS (IBM MODELS)

PROJECT DESCRIPTION :

**Fine tuning and Training of IBM Prithvi EO
Models for Earth Observation tasks.**

GROUP MEMBER :

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GROUP ID :

2026G264

GROUP GUIDE :

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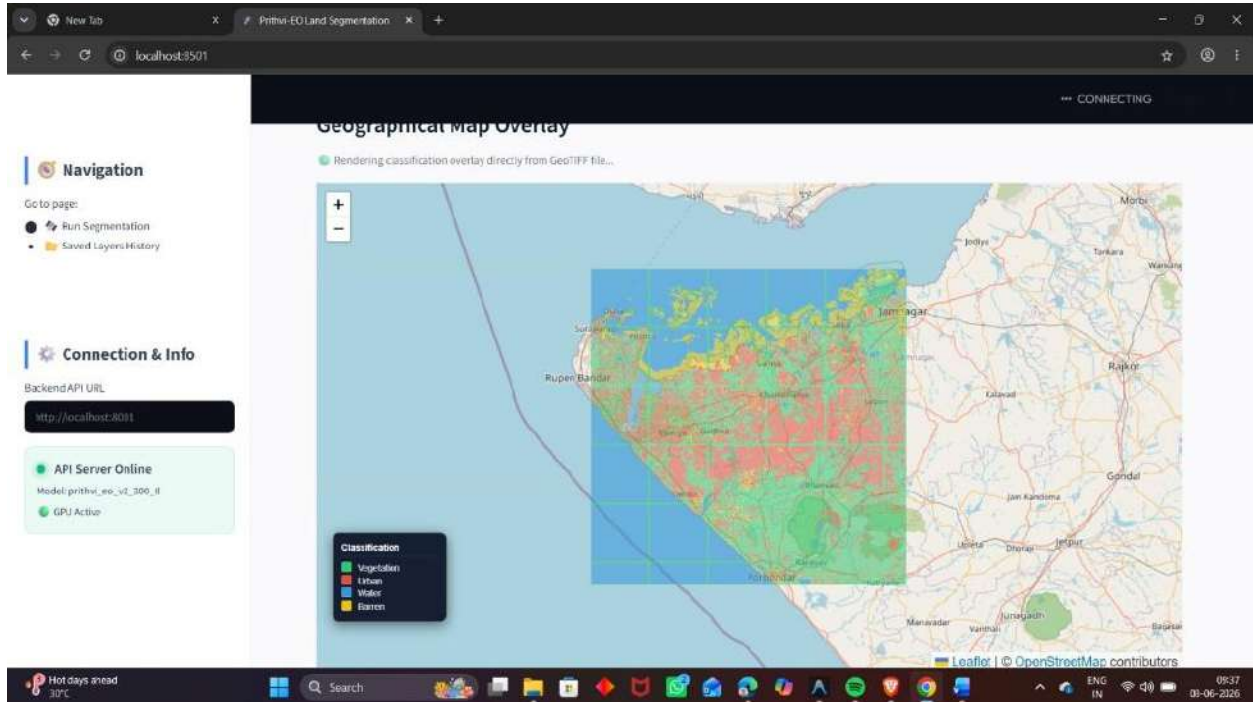
01/06/2026	Set up the workspace environment in WSL, configured project dependencies in requirements.txt, and initialized the FastAPI backend and Streamlit frontend application structure.
02/06/2026	Integrated the NASA-IBM Prithvi-300M foundation model and implemented the multi-band GeoTIFF band-merging logic (merge_bands) to support Sentinel-2, Landsat, and HLS input formats.
03/06/2026	Designed the interactive Streamlit dashboard layout, including the local filesystem browser, live execution logs, and inline active visualizer mapping.
04/06/2026	Integrated GeoServer publishing support via Web Map Service (WMS) protocol and implemented a local GeoTIFF base64-overlay fallback for offline map visualization.
05/06/2026	Overhauled the application styling to a premium enterprise-grade bright/light mode theme using slate panels, card layouts, and polished micro-animations.
06/06/2026	Designed and implemented a fragment-and-merge processing pipeline that splits massive 1GB+ Sentinel-2 inputs into 2048x2048 tiles to eliminate browser timeout and memory (OOM) errors.
07/06/2026	Holiday (Sunday)

Next Week Plan	Enhance the EO pipeline with new datasets, improved visualization, and testing for robust large-scale geospatial analysis.
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REFERENCE:

- ibm-nasa-geospatial (IBM-NASA Prithvi Models Family)
- Sentinel-2 | Copernicus Data Space Ecosystem
- <https://github.com/NASA-IMPACT/Prithvi-EO-2.0>
- <https://geoserver.org/>

SCREEN SHOTS:



Database Historical Records

Layer Name	COG Path / Created At	GeoServer WMS	Action
ahmedabad_sentinal_optimized	2026-06-07 13:08 /mnt/c/prithvi_datasets/cog_store/ahmedabad_sentinal_optimized_cog.tif	Local Only	
ahmedabad_sentinal	2026-06-07 12:37 /mnt/c/prithvi_datasets/cog_store/ahmedabad_sentinal_cog.tif	Local Only	
test_6	2026-06-05 12:17 /mnt/c/prithvi_datasets/cog_store/test_6_cog.tif	Published	
test_5	2026-06-05 12:11 /mnt/c/prithvi_datasets/cog_store/test_5_cog.tif	Local Only	
testing_3	2026-06-05 11:55 /mnt/c/prithvi_datasets/cog_store/testing_3_cog.tif	Local Only	
test_verify_stripe	2026-06-05 11:36 /mnt/c/prithvi_datasets/cog_store/test_verify_stripe_cog.tif	Local Only	

Hot days ahead 30°C

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